ProofOS

Production Operations Manual — iTechSmart ProofOS

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Version 1.0 · iTechSmart ProofOS

A multi-tenant RMM / IT-operations platform with a cryptographic proof ledger, AI copilot, ITSM, PAM, PSA, AIOps, remote control, and SSO.

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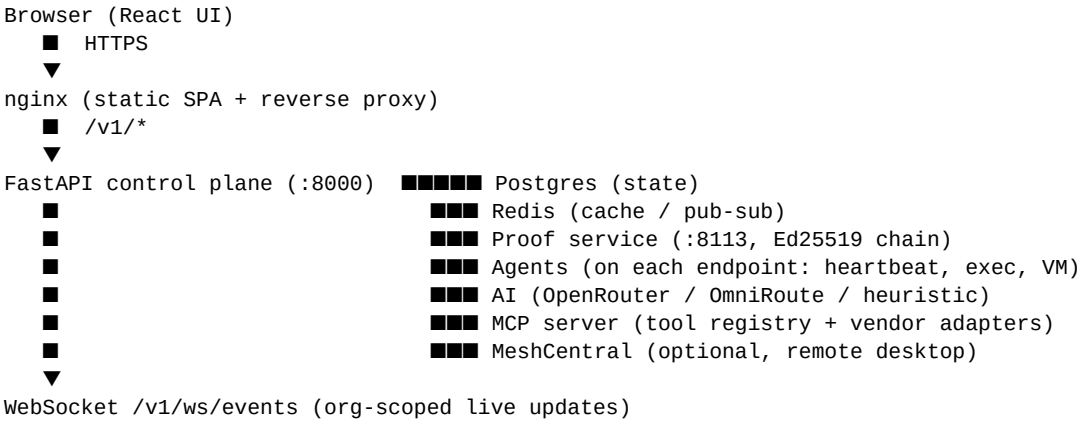
1. Introduction & Architecture

ProofOS consolidates RMM (remote monitoring & management), ITSM (ticketing, change, CMDB, knowledge), PAM (secrets vault), PSA (time tracking, consulting), and AIOps (anomaly detection, automation, AI copilot) into one pane of glass, with every action backed by a tamper-evident Ed25519 hash-chain proof ledger.

Components



Core loop: Detect → Decide → Do → Prove

- Detect — agents heartbeat (inventory + CPU/disk/mem) → status sync marks devices online/stale/offline → monitoring checks (ping/port/HTTP/SNMP) raise alerts.

• Decide — alerts/tickets auto-classified (P1–P4) → automations (trigger→condition→action) with autonomy routing (manual/semi/auto) + digital-twin simulation (blast radius + safe path).

• Do — human-gated approvals → remediation/command runs through the agent.

• Prove — every write emits a hash-chained Ed25519 receipt; verify/export/audit (CSV/JSON/PDF), with timestamp anchoring and iTechSmart ProofLink bridging.

Multi-tenant model

• An Org is a tenant — fully isolated (all data is org_id-scoped, including WebSocket events).

• Roles: admin, manager, engineer, auditor, viewer.

2. System Requirements

Tier	CPU	RAM	Disk	Notes
Minimum (dev / 1–5 clients)	2 vCPU	4 GB	10 GB SSD	single Compose host
Recommended (production)	4 vCPU	8 GB	50 GB SSD	+ managed Postgres/Redis optional
Agent (per endpoint)	~0.1 vCPU	~40 MB	—	heartbeat + command exec

Images: API ~250–350 MB · Web ~40–80 MB · Proof ~150 MB · Postgres 420 MB · Redis 61 MB · MeshCentral ~993 MB (optional).

3. Installation & Deployment

3.1 Prerequisites

• Docker Engine 24+ and Docker Compose v2

• A domain name and TLS certificate (recommended: Let's Encrypt)

• Ports: 80/443 (web), 8000 (API), 8113 (proof), 8096 (MeshCentral web), 44330 (MeshCentral agent)

3.2 Clone & configure

```
git clone <your-repo> proofos
cd proofos
copy .env.example .env
```

Edit .env (see [Configuration Reference](#4-configuration-reference)).

3.3 Start the stack

```
docker compose -f infra/docker-compose.yml up -d --build
```

Verify:

```
curl http://localhost:8000/health
# {"ok":true,"status":"ok","platform":"ProofOS","version":"1.0.0","db":"ok"}
```

Open the web UI at http://localhost:5173 (dev) or https://<your-domain> (production).

3.4 Production hardening (required before go-live)

• Put nginx (or Caddy/Cloudflare) in front with TLS termination.

- Set ENV=production (enables fail-closed validation — see §13).
- Use managed Postgres/Redis or enable host volume persistence (already in Compose).
- Set strong secrets: JWT_SECRET, PROOF_SIGNING_SEED_HEX, VAULT_KEY, AGENT_TOKEN.
- Set ALLOW_BOOTSTRAP=false after creating your first org.
- Back up Postgres + the proof chain (see §12).

4. Configuration Reference

All settings are read from environment variables (via .env).

Core

Variable	Default	Purpose
DATABASE_URL	postgresql+psycopg2://…	SQLAlchemy DB URL (or sqlite:///./app.db)
REDIS_URL	redis://localhost:6379/0	cache / pub-sub
ENV	development	production enables fail-closed checks
JWT_SECRET	—	required, ≥32 chars
PROOF_SIGNING_SEED_HEX	—	64 hex chars, Ed25519 signing seed
AGENT_TOKEN	—	shared agent auth (optional; per-device secrets preferred)
ALLOW_BOOTSTRAP	true	allow first-run bootstrap (disable in prod)
WEB_URL	http://localhost:5173	SPA origin (SSO redirects)
APM_ENABLED	false	record request-trace spans

AI

Variable	Default	Purpose
AI_PROVIDER	heuristic	heuristic · omniroute · openrouter
OMNITROUTE_URL / OMNITROUTE_API_KEY	—	OmniRoute gateway
OPENROUTER_API_KEY / OPENROUTER_MODEL	—	OpenRouter (free models)

Notifications

Variable	Purpose

SLACK_WEBHOOK_URL	Slack alerts
SMTP_HOST / SMTP_PORT / SMTP_USER / SMTP_PASSWORD / SMTP_FROM / SMTP_TO	email alerts
ALERT_WEBHOOK_URL	generic webhook alerts

Security / Vault

Variable	Purpose
VAULT_KEY	Fernet key for secrets vault (auto-derived from JWT_SECRET if unset)

Integrations

Variable	Purpose
ITECHSMART_MCP_URL	iTechSmart MCP server (http://127.0.0.1:3200)
ITECHSMART_API_KEY	iTechSmart MCP key (x-itechsmart-mcp-key header)
MESHCENTRAL_URL	MeshCentral server (https://localhost:8096)
VNC_PORT	VNC launch port (5900)
FCM_PROJECT_ID / FCM_SERVICE_ACCOUNT_JSON	Firebase Cloud Messaging

SSO

Variable	Purpose
GOOGLE_CLIENT_ID / GOOGLE_CLIENT_SECRET / GOOGLE_REDIRECT_URI	Google OIDC
SAML_ENTITY_ID / SAML_ACS_URL / SAML_IDP_METADATA_URL	Okta/SAML

5. First Run & Bootstrap

• Open the web UI → the login screen shows "Sign in", "Sign in with Google", and the bootstrap hint.

• Enter an Org name (e.g. Acme Corp) and click Sign in.

• On first run, the app auto-bootstraps: creates the org + an admin user (admin-<id>@proofos.dev) and logs you in.

• Immediately: change the admin password and email (Admin → Users), then set ALLOW_BOOTSTRAP=false in .env and restart.

API alternative: POST /v1/auth/bootstrap {"name": "Acme Corp"} → returns {org, user, access_token}. Idempotent — safe to re-run.

6. Multi-Tenancy & User Management

6.1 Create additional orgs (tenants)

Each client = one Org. Via API:

```
curl -X POST http://localhost:8000/v1/auth/orgs -H "Authorization: Bearer $JWT" -H "Content-Type: applica
```

6.2 Manage users

Admin → Users panel, or:

```
# list
curl "http://localhost:8000/v1/auth/users?org_id=1" -H "Authorization: Bearer $JWT"
# create
curl -X POST http://localhost:8000/v1/auth/users -H "Authorization: Bearer $JWT" -H "Content-Type: applic
```

Roles:

- admin — full control (users, approvals, secrets, SSO, VM control)
- manager — operations, approvals, VM control
- engineer — devices, tickets, scripts, monitoring
- auditor — read-only + audit
- viewer — read-only

All routes are org_id-scoped: a user of Org 1 cannot see or touch Org 2 data.

7. Device Enrollment & Agent

7.1 Enroll a device (UI)

- Devices → fill hostname, os, ip, select device_type → Register.
- A one-time enrollment secret is shown (copy it now — it's shown once).

7.2 Install the agent

```
python agent.py --api http://<proofos-host>:8000 --device-id <id> --token <device_secret>
```

The agent then:

- Heartbeats inventory + telemetry (CPU/disk/mem) periodically
- Polls for commands and executes them (shell/PowerShell)
- Runs VM/hypervisor commands (see §11)

7.3 Device states

Status is derived from last_seen:

- online — heartbeat < 180 s
- stale — 180–600 s
- offline — > 600 s (raises a critical alert)

7.4 Device types

workstation, server, access_point, gateway, switch, node.

7.5 Remote command

Devices → click a device → type a command → Run. Output appears in run history (proof-logged).

8. Feature Walkthroughs

8.1 Dashboard

- Exec view (KPIs: devices, tickets, proof records, alerts, approvals, audit score) and Operations view (device health, tickets by status, alerts, devices by type).
- Quick actions: Run audit, Add device, New ticket, Run checks.
- Live WebSocket updates (no refresh needed).

8.2 Devices

Register → search → filter by type → open detail (telemetry, last seen, rotate secret, delete) → Refresh statuses.

8.3 Tickets

- Tickets → title + details + priority (auto-severity P1–P4 by SLA).
- Click a ticket → Status transitions (open → in_progress → resolved → closed), add Comments.
- SLA due/breach tracking with due_at.

8.4 Monitoring

- Monitor → select device → kind (ping/port/http/snmp) → target + threshold → Add check.
- Run checks for all devices → raises alerts on failure.
- Alerts → ACK, and auto-delivered via Slack/email/webhook/FCM (see §9).

8.5 Scripts

- Scripts → name + body + version → Create script.
- Select a script + a device → Run → output in Run history (exit code, stdout).

8.6 Automations & Approvals

- Automations → name, trigger, condition JSON, action, autonomy (manual/semi/auto), optional remediation script → Create rule.
- Simulate → digital-twin preview: blast radius + safe path.
- When triggered, routing follows autonomy:
- manual → creates an Approval (human must approve/deny)
- semi → executes but flags for review
- auto → executes + logs
- Admin → Approvals → Approve / Deny (every decision proof-logged).

8.7 Proof Ledger

- Proof view: list records, Verify (checks hash-chain integrity), Export (CSV/JSON), Compliance report (PDF).
- Optional timestamp anchor (local SHA-256 commitment) and iTechSmart ProofLink bridge (seal receipts).

8.8 AI Copilot

• AI Copilot → type a question (e.g. "offline devices") → Ask → risk/triage answer over live state.
• Propose a remediation → generates a human-gated approval.

8.9 Admin

• Approvals, Users (add/roles), Push notifications (Firebase FCM — register token + send test), Inventory by type.

8.10 CMDB

• CMDB → name + type + attributes JSON → Add CI. Configuration items: server, switch, application, service, database, network.

8.11 Knowledge Base

• Knowledge → title + body + category + tags → Add article. Search across title/body/tags.

8.12 Secrets Vault (PAM)

• Vault → name + value + category → Store secret (encrypted with Fernet at rest).
• Reveal decrypts and is audit-logged (proof receipt).

8.13 Change Management

• Changes → title + description + risk + impact + plan → Create change.
• Lifecycle: draft → Submit → Approve → implement → close.

8.14 Time Tracking (PSA)

• Time → minutes + notes + billable → Log time. Summary: total/billable hours.

8.15 Audits & PDF Reports

• Audit → kind (security/compliance/governance/full) → Run audit → scored findings → Download PDF.

8.16 Onboarding

• Onboarding → Init checklist → toggle steps as done (client onboarding walkthrough).

8.17 Consulting

• Consulting → save Client profile → create Projects (planning/active/on-hold/done).

8.18 Patches

• Patches → name + device + category + severity → Register patch → Deploy (runs winget/apt/yum via the agent).

8.19 VMs

• VMs → name + host + hypervisor (hyper-v/libvirt/vmware/qemu) → Register VM → Start/Stop/Restart (native hypervisor commands via the agent).

9. Integrations

9.1 iTechSmart (UAIO / ProofLink / MCP)

Configure ITECHSMART_MCP_URL + ITECHSMART_API_KEY, then:

- GET /v1/itechsmart/status — live UAIO health
- GET /v1/itechsmart/tools — 66 MCP tools
- GET /v1/itechsmart/chain — ProofLink Plus receipts
- POST /v1/itechsmart/mcp/call — invoke any tool (e.g. query_uaio_status)
- POST /v1/itechsmart/seal — seal a ProofOS receipt into ProofLink Plus

9.2 MCP (ProofOS as an MCP server)

- GET /v1/mcp/tools — registry of 25+ tools (device.*, ticket.*, proof.verify, audit.run, vm.*, approval.*, itechsmart.mcp_call, …)
- GET /.well-known/agent.json — A2A agent card for agent discovery.
- Connect Claude/Cursor/LangGraph/CrewAI to ProofOS as a tool.

9.3 Notifications

- Slack: set SLACK_WEBHOOK_URL.
- Email: set SMTP_*.
- Generic webhook: set ALERT_WEBHOOK_URL.
- Firebase push: set FCM_* and register device tokens (Admin → Push).

9.4 Metrics ingestion (Datadog/Prometheus/Grafana)

POST /v1/integrations/metrics — push custom metric points; they feed the anomaly detection.

10. Single Sign-On (SSO)

10.1 Google (OIDC)

- Google Cloud Console → Credentials → OAuth client ID (Web).
- Redirect URI: https://<host>/v1/sso/google/callback.
- Set GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, GOOGLE_REDIRECT_URI, WEB_URL.
- Ensure the user's Google email matches a ProofOS user → click Sign in with Google.

10.2 Okta / SAML 2.0

- Okta → Applications → Create App Integration → SAML 2.0.
- SSO URL: https://<host>/v1/sso/acs; Audience URI: https://<host>/sp.
- Add email attribute → user.email, Name ID = EmailAddress.
- Copy the IdP metadata URL → set SAML_IDP_METADATA_URL.
- GET /v1/sso/sp-metadata returns ProofOS's SP metadata for Okta.

Full steps: docs/OKTA.md.

11. Remote Control & Virtualization

11.1 MeshCentral (open-source remote desktop)

MeshCentral ships in the Compose stack (port 8096). To activate:

- Open https://<host>:8096 → create the first account (becomes admin).
- In MeshCentral → Add Agent → install on each endpoint.
- ProofOS Devices → Open KVM (VNC) returns a connect link (or use the MeshCentral deep-link).

11.2 VNC (Linux)

- Devices → Open KVM (VNC) queues x11vnc on the agent and returns vnc://<ip>:5900.

11.3 Hypervisors

VM control commands are generated per hypervisor and run through the agent:

- Hyper-V: Start-VM / Stop-VM / Restart-VM
- libvirt/QEMU: virsh start|shutdown|reboot
- VMware: vim-cmd vmsvc/power.on|off|reboot

12. Operations Runbook

12.1 Backup

- Postgres: pg_dump (or your managed provider's backups).
- Proof chain: the proof_records table (in Postgres) + export via /v1/proof/export.
- Secrets vault: encrypted at rest; back up VAULT_KEY separately (off-box, e.g. age-encrypted).

12.2 Monitoring the platform

- GET /health — liveness.
- GET /v1/analytics/apm — request latency (enable APM_ENABLED=true).
- Container healthchecks are defined in Compose.

12.3 Scaling

- Vertical: raise CPU/RAM (see §2).
- Horizontal: run multiple API workers behind a load balancer; use managed Postgres/Redis; move the proof-chain head to Postgres.

12.4 TLS

Terminate TLS at nginx/Caddy/Cloudflare. Keep :8000, :8113, :8096 internal (not public).

13. Security Hardening

- ENV=production — fails closed on missing critical secrets.
- ALLOW_BOOTSTRAP=false after first org.

- AGENT_TOKEN or per-device secrets (enforced) — no open heartbeat endpoint.
- VAULT_KEY in an encrypted secrets manager, never in plaintext VCS.
- Rate-limited login (sliding window) + security headers are enabled by default.
- Rotate JWT_SECRET / PROOF_SIGNING_SEED_HEX only with a planned re-key (proof chain depends on the seed).
- Restrict the public surface to 443 (web) only.

14. Troubleshooting

Symptom	Cause / Fix
403 org scope mismatch	Wrong org_id or role; use your own org id
401 unauthorized on iTechSmart seal	ProofLink Plus has separate auth; MCP key only covers MCP tools
Device stuck offline	Agent not heartbeating; check agent.py connectivity + token
Proof verify fails	Chain tampered or re-seeded; restore from backup
SQLite bad parameter (dev)	Use Postgres in production; SQLite pool fixed for concurrent reads
SSO login loops	Check WEB_URL + redirect URI match exactly

Logs: docker compose logs -f api web proof.

15. API Reference

All routes are under /v1/*, JWT via Authorization: Bearer <token>. Interactive docs at GET /docs (Swagger).

Prefix	Domain
/v1/auth	login, bootstrap, orgs, users, SSO
/v1/devices	enrollment, heartbeat, telemetry, commands
/v1/tickets	tickets + comments
/v1/monitor	checks, alerts, ack
/v1/scripts	scripts + runs
/v1/automations	rules + simulate
/v1/approvals	human-gated approvals
/v1/ai	ask, propose
/v1/proof	verify, export, report, anchor
/v1/audits	run + findings
/v1/cmdb, /v1/knowledge, /v1/secrets	CMDB / KB / vault
/v1/changes, /v1/time	change mgmt / time
/v1/vms, /v1/remote, /v1/patches	virtualization / remote / patches
/v1/analytics	anomalies, root-cause, APM

/v1/integrations	metrics ingestion
/v1/itechsmart, /v1/mcp, /v1/push, /v1/sso	integrations / MCP / push / SSO
/v1/ws/events	WebSocket (org-scoped)

16. FAQ

Q: One platform or many? One — ProofOS replaces RMM + ticketing + PAM + PSA + AIOps + audit in a single multi-tenant pane.

Q: How do I add a second client? Create a new Org (tenant); its data and users are fully isolated.

Q: Does the proof chain survive restarts? Yes — records persist in Postgres; verify recomputes the chain.

Q: Can I run the agent on Windows? Yes — the agent runs Python; Windows devices use PowerShell for commands and Hyper-V for VM control.

Q: Is the Bitcoin anchor on-chain? The built-in anchor is a local SHA-256 commitment; on-chain anchoring is available via iTechSmart ProofLink Plus.

Q: Where are passwords stored? Passwords are bcrypt-hashed; vault secrets are Fernet-encrypted.

ProofOS — secure by design, provable by default.